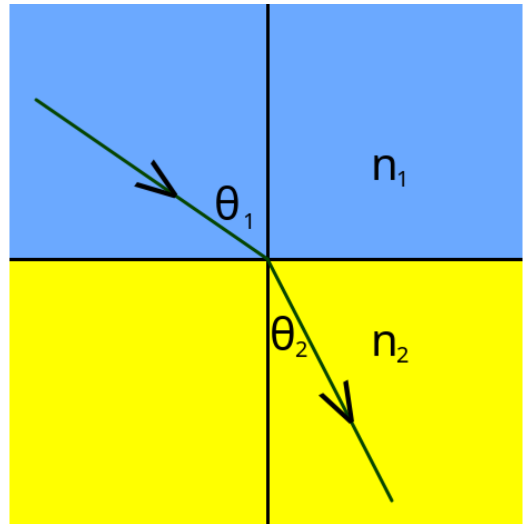
Example Lab Proposal: Snell's Law
PH150

Student Name
Date

Introduction to Snell's Law



Snell's law is a law of optics involving the relationship between the angle of incidence and the angle of refraction when light travels from one medium to another. The formula is $n_1 \sin\theta_1 = n_2 \sin\theta_2$, where n_1 is the incident index, n_2 is the refracted index, θ_1 is the incident angle, and θ_2 is the refracted angle. n is calculated by the formula $n = c/v$ where c is the speed of light in a vacuum, and v is the speed of light in the medium in question. However, for the purposes of this experiment, we will not be calculating n using the speed of light; we will be calculating n based on a combination of Snell's Law and commonly accepted values of n.

Procedure, Proposed Analysis, and Expected Results

Procedure

For this experiment, we will measure the index of refraction for three selected mediums, glass, acrylic, and water, by shining a laser through air into each of our second mediums at various angles, measuring the angles of incident and refraction, and then plugging those values into Snell's law to calculate the refraction index of the respective second medium. We will be measuring the angles by holding a 360° protractor by hand up to each of the mediums while shining the laser through (with the center of the protractor aligned with where the laser goes from one medium to another), and thus we will have a fair amount of uncertainty. Because of this anticipated uncertainty, for now I will say it is safe to approximate the index of incidence of air to be 1, and prove it later on in this proposal. It's commonly accepted value is 1.0003, however including that much precision will be negated when our uncertainty for the measurement of the angles are calculated.

Proposed Analysis & calculating expected uncertainty

The commonly accepted indices of refraction of glass, acrylic, and water are 1.52, 1.49, and 1.333, respectively. So we should expect to get values relatively close to that. However, how much uncertainty should be expected? Let's experiment with a couple of arbitrary test values for θ_1 and θ_2; let's say 60° and 30° respectively. A reasonable expected amount of error for a singular measurement of each of the angles might be around plus or minus one degree. So if we take those selected arbitrary test values for θ_1 and θ_2 and plug them into Snell's Law, approximating n_1 to be 1, then we get $n_2 = 1.73205$. If we change θ_1 to 59° and also change θ_2 to 31° (varying our measurement by our expected uncertainty of plus or minus one degree), then our value of n_2 changes to 1.66428. Likewise, if we e change θ_1 to 61° and change θ_2 to 29°, then our value of n_2 changes to 1.80405. If we find half the difference (approximately the same as the difference between $n_2 = 1.73205$ and our other two calculated values) between each of those values for n_2, then we get an uncertainty of plus or minus 0.069885, or approximately 0.070.

Proving that we can negate the precision of the incident index of air

Now to prove that we can negate the precision on the index of air, 1.0003. Using those same angles, 60° and 30°, but this time $n_1 = 1.0003$ instead of 1, we get that $n_2 = 1.73257$, which is a difference of 0.00052 from our value of n_2 from when we approximated the index incidence of air to be 1. That difference of 0.00052 is well within the range of our expected uncertainty of 0.070, thus proving that the precision of using 1.0003 for the index incidence of air to be 1.

We will measure eight different angles in for each medium, then from each of those eight angles, we will calculate both the index of refraction for that medium, and the uncertainty in our measurement. Our measurements will be plugged into a table like this:

Incident Angles	**Acrylic Refraction Angle**	**Glass Refraction Angle**	**Water Refraction Angle**
10°			
20°			
30°			
40°			
50°			
60°			
70°			
80°			

Anticipated Difficulties

- Holding the laser still long enough for someone else to acquire a measurement of both angles with the protractor. One remedy might be a mount which we can attach the laser to.

Preliminary List of Equipment Needed

- Laser
- Something to mount the laser on, to prevent it from moving
- Clear acrylic block
- Clear glass block
- Clear cryolite container for water (cryolite has a really close index of refraction to water)
- Water to put in previously-mentioned clear cryolite container
- 360° protractor